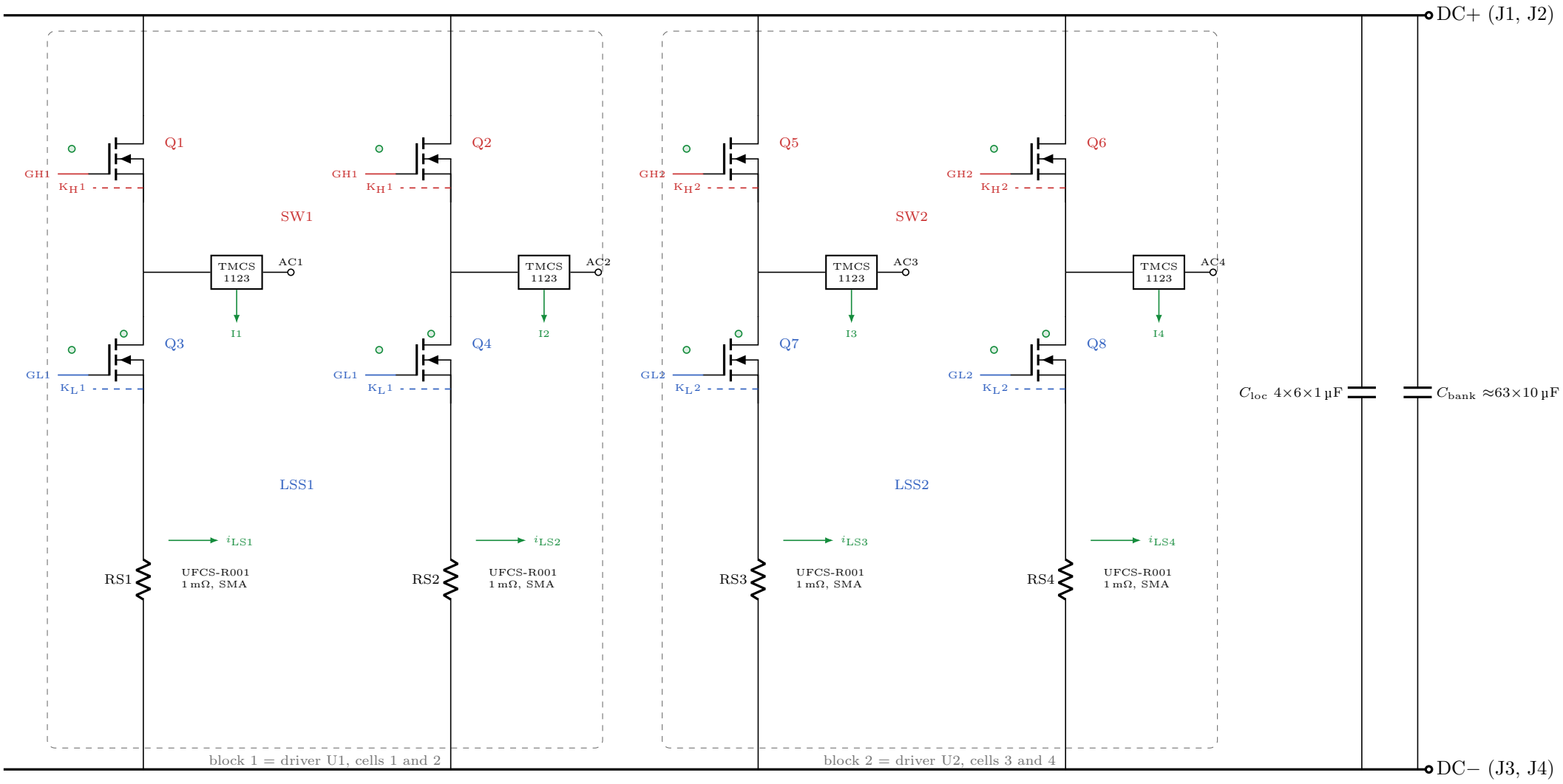
(a) Power stage: one leg, 4× EPC2361 per switch, one cell per transistor pair

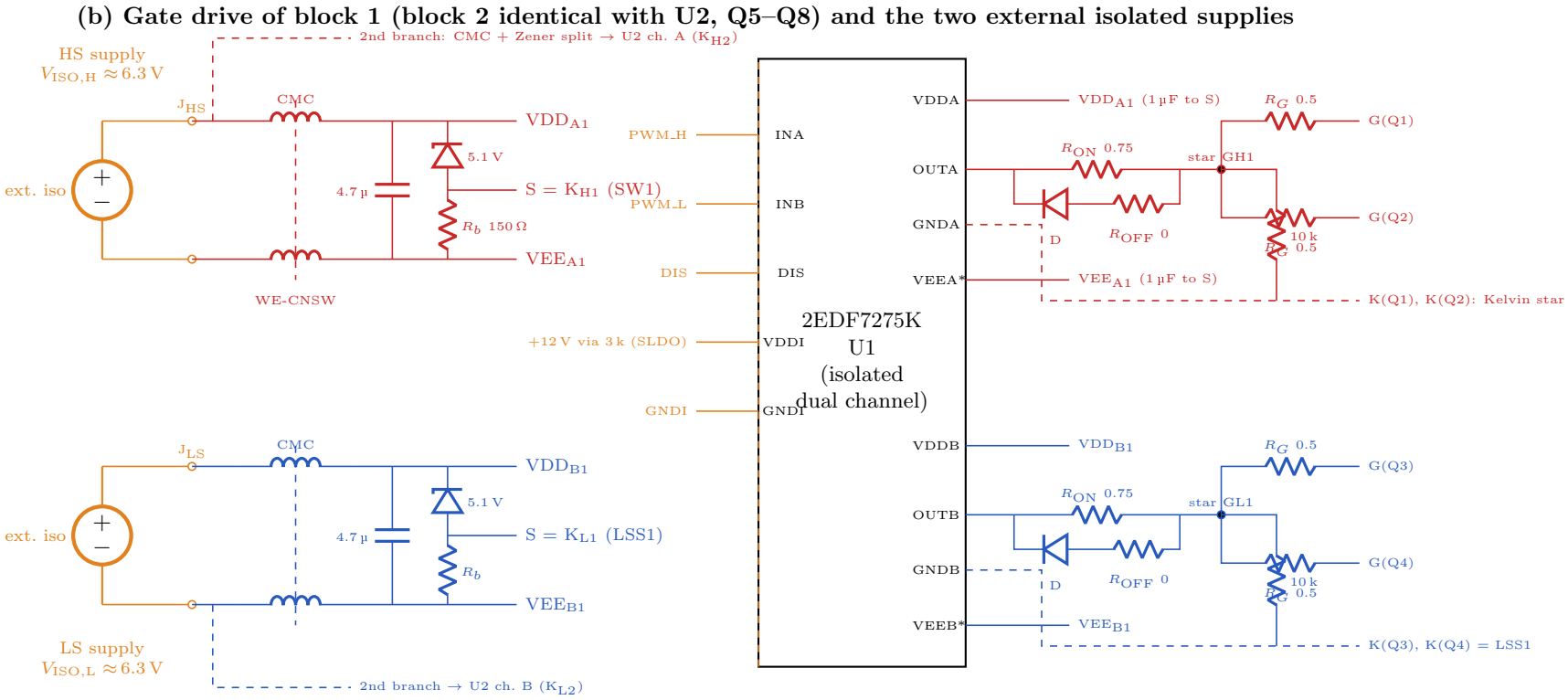


- U.FL (bottom side, at the device): V_{GS} of every HS and LS FET, V_{DS} of every LS FET, with series-R footprint (0 Ω or 953 Ω = 20:1 into 50 Ω)

Two cells of a block share one driver channel per switch (star + per-FET R_G); the LS sources of a block meet only through Kelvin traces, so each shunt measures its own cell.

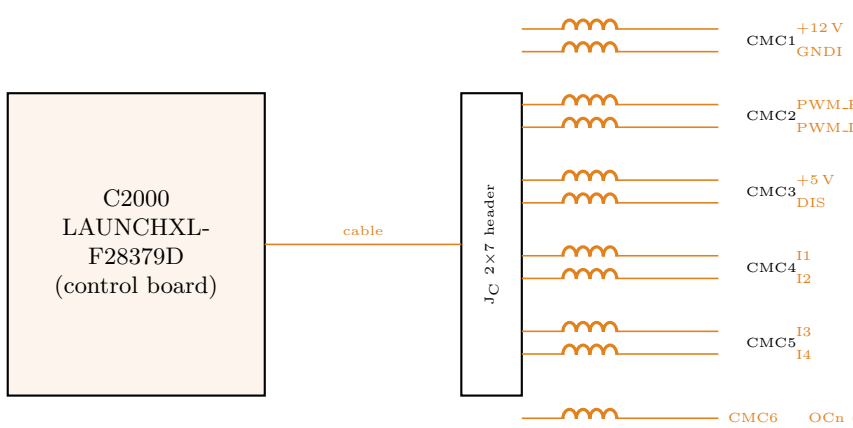
AC1–AC4 are joined off the board (busbar to the load); I1–I4 summed in the controller give the leg current.

(b) Gate drive of block 1 (block 2 identical with U2, Q5–Q8) and the two external isolated supplies



* driver VEE pin = negative rail; GND pin = Kelvin source of the pair

(c) Control interface: one cable connector with common-mode chokes to the C2000 (TMS320F28379D)



+12 V: driver input side (VDDI via SLDO) and the 12 V fan
+5 V: TMCS1123 supply (4×14 mA)
PWM_H / PWM_L: fanned out to U1 and U2 (47 Ω + 100 pF each, 10 k pull-downs)
DIS: both drivers
I1–I4: 2.5 V ± 25 mV/A (±96 A) → scale to the 3.3 V ADC on the control board
Gate supplies: two external isolated supplies (HS, LS), not on this cable